



Jarin Tasnim Ishika

PhD Candidate, Department of Computer Science and Engineering
University of North Texas

About

I am a PhD Candidate at the University of North Texas. My current advisor is Dr. Stephanie Ludi, and I work in the DiscoverABILITY research lab. My research spans Software Engineering and Software Development Tools, with a focus on Software Accessibility, Computer Science Education, Human–Computer Interaction, and Cognitive Science. Beyond my core areas, I am also interested in Natural Language Processing, Human–Machine Intelligence, and Information Retrieval.

Research

2025 — Ongoing Investigating usability challenges faced by ADHD computer science students and software engineering professionals while using IDEs in text-based programming

Education

Aug 2023 — Now **PhD in Computer Science and Engineering**
University of North Texas, Denton, TX · GPA 3.944 / 4.0

2015 — 2019 **BSc in Information Technology**
Jahangirnagar University, Dhaka, Bangladesh · GPA 3.74 / 4.0

Research Experience

Aug 2023 — Now **PhD Candidate — DiscoverABILITY Lab**
University of North Texas, advised by Dr. Stephanie Ludi. Served as Graduate Research Assistant, May – Aug 2025.

Teaching Experience

Aug 2025 — Now **Graduate Teaching Assistant — CSCE 4901: Software Development Capstone I**
Supervised all project teams on multi-component software design, providing guidance and tracking progress across the semester.

Jan 2024 — Apr 2025 **Graduate Teaching Assistant — CSCE 1035: Computer Science 2**
Python programming course. Supported instructor operations and evaluated assignments for roughly 80 students.

May — Jul 2024 **Graduate Teaching Assistant — CSCE 5465: Usability Testing**
Ten-week usability-focused project course. Oversaw project development and monitored progress throughout.

Aug — Dec 2023 **Graduate Teaching Assistant — CSCE 1030: Computer Science 1**
Led a nine-hour weekly lab for about 90 students on C and C++ fundamentals.

Industry Experience

Jan — Jun 2020

Software Engineer Intern

Board of Intermediate and Secondary Education, Dhaka, Bangladesh — authored technical design documents, tracked client feature requests and demo feedback, and ran unit and system testing.

Projects

Graduate

A Framework Using Large Language Models to Transform Developer Suggestions into Structured Questions for Evaluating Program Comprehension

Developers spend 50–70% of their time comprehending code, yet the questions they ask while exploring an unfamiliar codebase are informal and unstructured. This framework uses large language models to convert developer suggestions gathered from social media into structured questions that assess program comprehension against defined metrics, validated across three open-source LLMs.

<https://github.com/JarinTasnimIshika121/A-Framework-Using-Large-Language-Models-to-Transform-Developer-Suggestions-into-Structured-Questions->

Graduate

A Comprehensive Survey of Single-Source Shortest Path Algorithms: Evolution and Optimization

Survey tracing how the long-standing $O(m + n \log n)$ sorting barrier for single-source shortest paths was broken — from the 2023 randomized bundled-Dijkstra method for undirected graphs to a 2025 deterministic algorithm using pivot-based recursive decomposition, reaching $O(m \log^{2/3} n)$.

<https://github.com/JarinTasnimIshika121/A-Comprehensive-Survey-of-Single-Source-Shortest-Path-Algorithms-Evolution-and-Optimization->

Graduate

Text Retrieval System

Information retrieval system built on vector space models and term weighting, ranking documents by cosine similarity between query and document vectors, with Porter stemming for term normalisation.

<https://github.com/JarinTasnimIshika121/Text-Retrieval-System->

Undergraduate

OMR Sheet Reading System

Image-processing pipeline in Java that evaluates optical mark recognition sheets, using edge detection and perspective transformation to read and grade answer forms automatically.

<https://github.com/IshikaTasnim/OMR-Sheet-Reading-System-using-Java>

Undergraduate

Online Examination System

Web application giving teachers an environment to set tests and proctor them remotely.

<https://github.com/IshikaTasnim/Online-Examination-System>

Undergraduate

Egg Catcher

Graphical game built with java.awt and java.io — an academic project that gave me early insight into how users respond to real-time interaction.

<https://github.com/IshikaTasnim/Egg-Catcher-Graphical-Computer-Game>

Graduate Coursework

Advanced Topics in Human–Machine Intelligence · Advanced Research in HCI · Usability Testing · Software Engineering · Advanced Software Engineering · Software Development for AI · Machine Learning · Information Retrieval · Algorithm Analysis

Skills

Languages	C / C++ · Java · Python · JavaScript · PHP · HTML · CSS · Assembly · LaTeX
Tools	Git · TensorFlow · Node.js · PostgreSQL · MySQL · MATLAB · Jupyter · Eclipse · Android Studio
Research Tools	Qualtrics · ATLAS.ti
Methods	Survey Design · Interview Design · Qualitative Data Analysis · Quantitative Data Analysis · System Design Thinking
Platforms	Unix / Linux · Windows · Android

Contact

Email	jarintasnimishika@my.unt.edu
LinkedIn	https://www.linkedin.com/in/jarin-tasnim-ishika-916508240/
GitHub	https://github.com/JarinTasnimIshika121
Substack	https://substack.com/@jarintasnimishika